

PARKINSON'S DISEASE

Physiotherapy Assessment & Management

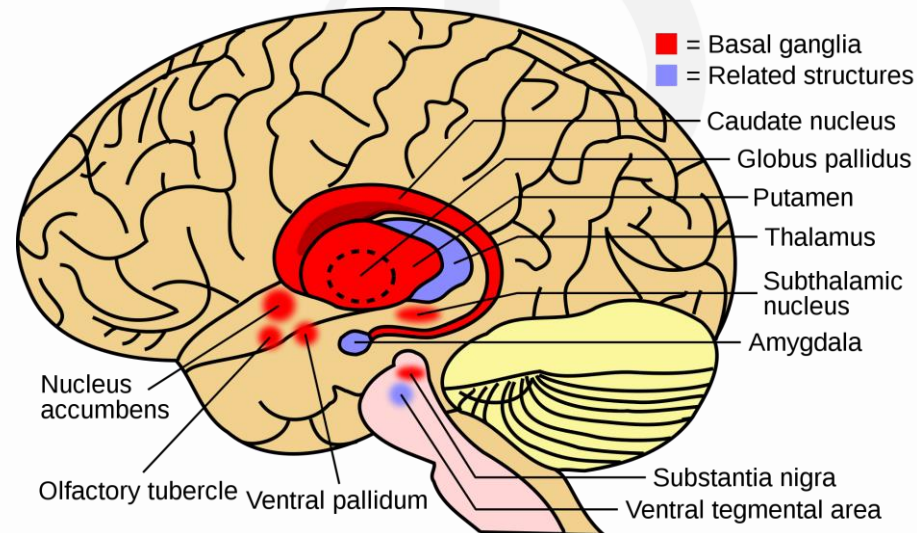
Introduction :

- Progressive disorder of Central Nervous System withy both motor and non-motor symptoms
- It shows a group of symptoms
- Motor symptoms include
 - Tremor
 - Bradykinesia
 - Postural instability
- Non-motor symptoms include
 - Loss of sense of smell
 - Constipation
 - Mood disorder
 - Excess salivation
- Onset is with a slow rate of progression



Introduction :

- There is the degeneration of the dopaminergic neurons in the basal ganglia in the **Pars Compacta** of the **Substantia Nigra**.
- Substantia Nigra is one of the parts of the Basal Ganglia
- This is divided into two parts – Dorsal Part (Pars Compacta) & Ventral Part (Pars Reticulata)
- The dorsal part of the substantia nigra contains dopaminergic neurons that project into Corpus Striatum (Putamen & Caudate Nucleus)



Etiology :

- Idiopathic
- Multi-Infarct Disease
- Infection
- Toxicity
- Drug toxicity
- Alzheimer's Disease
- Shy-Drager Syndrome
- Wilson's Disease
- Abnormal calcium deposition in basal ganglia
- Traumatic (Eg – Punch Boxers Syndrome)



Clinical Features :

- **Bradykinesia**
 - Slowing of movement with decrease in amplitude and intensity of contraction
 - There is time lag between patient's desire to act and final action
- **Rigidity**
 - Muscle stiffness
 - May involve all body parts
- **Tremor**
 - Initial symptom
 - Resting tremor is seen
 - Commonly seen in the limbs but sometimes seen in lips and tongue
- **Posture**
 - Flexed posture of the trunk and limbs together with difficulty in balancing

Clinical Features :

- **Gait**
 - Initially there will be lack of associated movements like arm swing during walking
 - Gradually, gait becomes slow with shuffling and smaller steps
 - Initiation of walking is difficult
 - After initiation, patient walks fast with smaller steps in a bend position because the COG shifts forward.
 - This is called Festinating Gait.
 - Patient may get freeze during the initiation of the gait or during any turn while walking
 - Some patients may have difficulty in stopping suddenly and needs external force to start or stop the gait

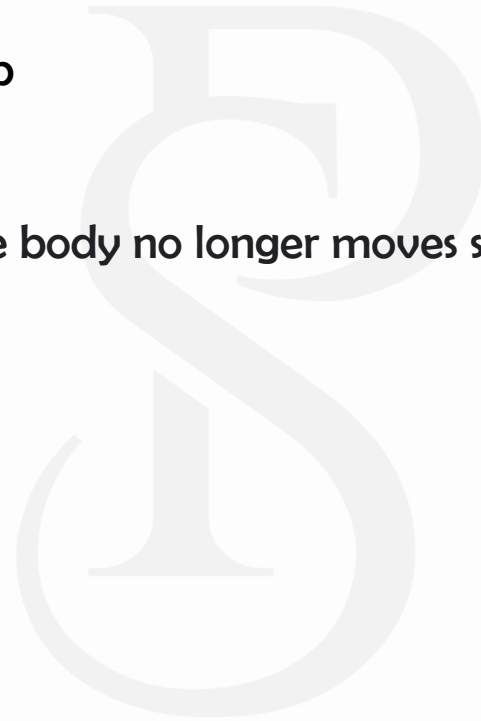
Clinical Features :

- Facial Attitude
 - Expressionless face
 - The person appears to be continuously staring
 - Increased salivation
 - Shiny skin
- Speech
 - Slurred speech
 - Handwriting abnormality



Clinical Features :

- Pill-Rolling movement of Hands
 - Circular movement of the thumb
- Cogwheel Rigidity
 - The muscles become stiff and the body no longer moves smoothly. When movement becomes jerky, it is called cogwheel rigidity



Assessment



Observation :

- General condition of the patient
- Patient's behavior
- Patient's level of awareness
- Posture
- Injury
- Skin
- Gait pattern
- Tremor
- Any deformity



Examination :

- GCS
- Memory examination
- Speech examination
- Sensory Examination
 - **Superficial** – Pain, Touch, Temperature
 - **Deep** – Pressure, Vibration, Joint position sense
 - **Combined cortical sensation**
 - Tactile localization
 - Stereognosis
 - Two point discrimination
 - Double simultaneous stimulation

Examination :

- Cognitive Function
 - Memory, Orientation, Conceptual reasoning, Judgement should be checked
 - Done through Mini Mental Status Examination (MMSE)
 - Set of 11 questions used to examine the cognitive function like :
 - Thinking
 - Communication
 - Understanding
 - Memory

Examination :

- Mini Mental Status Examination

1. Time, Day, Date, Month, Year – 5
2. Place, Floor, City, State, Country - 5
3. Name 3 objects and ask to repeat – 3
4. Serial subtraction (100-7-7-7) – 5
5. Recall 3 objects named above – 3
6. Name two words – 2
7. Follow 3 step command – 3
8. Copy intersecting pentagons – 1
9. Repeat therapist line – 1
10. Follow written command 1
11. Write a sentence

Examination :

- Postural control and balance
 - Posture instability is seen in the patients of the Parkinson
 - Done through **Berg Balance Scale**
 - Set of 14 activities used to examine the postural control and balance.
 - Each activity carries 5 points ranging from 0 to 4

Examination :

- **Berg Balance Scale**

- Sitting to Standing
- Standing unsupported
- Sitting unsupported
- Standing to sitting
- Transfers
- Standing with eyes closed
- Standing with feet together
- Reaching forward with outstretched arm
- Retrieving object from floor
- Turning to look behind
- Placing alternate foot on stool



Examination :

- Chest Assessment
 - Breathing Pattern
 - Inspiratory – Expiratory Ratio
 - Chest wall mobility
- Gait Parameters
 - All the gait parameters should be examined
- Coordination
 - Finger – Nose – Finger Test
 - Heel shin test
 - Tandem Walking
- Coordination
 - Checked by slow passive movement. Severity of Rigidity can be validated by the level of resistance

Hoehn and Yahr Staging of Parkinson :

- **Stage 1**
 - Sign & Symptoms on only one side
 - Mild symptoms
 - Symptoms inconvenient but not disabling
 - Tremor of one limb
- **Stage 2**
 - B/L involvement
 - Minimal disability
 - Normal Balance
- **Stage 3**
 - B/L involvement with mild to moderate disability
 - Mild postural imbalance but patient leads independent life



Hoehn and Yahr Staging of Parkinson :

- Stage 4
 - B/L involvement
 - Postural instability
 - Able to walk or stand unassisted
- Stage 5
 - Severe
 - Patient not able to walk or stand
 - Completely restricted to bed or wheelchair

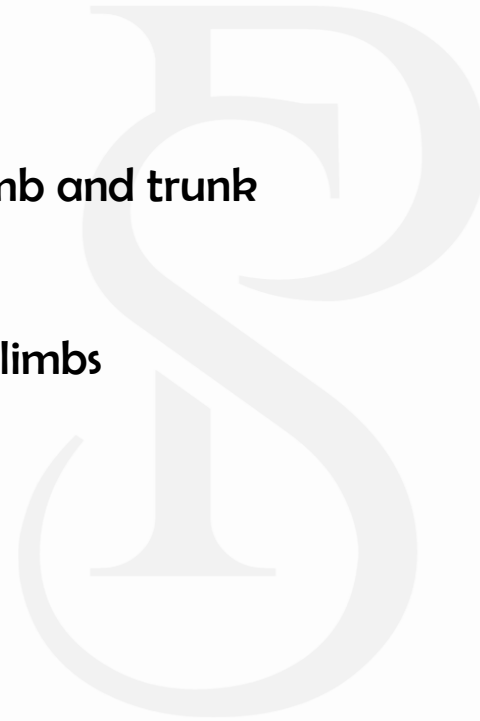
Physiotherapy Management

Aim of Treatment :

- Maintain and improve levels of function and independence
- Improve mobility
- Correct and improve abnormal movement patterns and posture
- Maximize muscle strength and joint flexibility
- Correct and improve balance and minimize the risk of falls
- Maintain a good breathing pattern
- Enhance the effects of drug therapy
- Improve ADL

PT Management :

- **Mobility**
 - Gentle rocking exercise
 - Rotational movement of limb and trunk
 - Relaxation techniques
 - Gentle ROM exercise of the limbs
 - Stretching
 - Positioning
 - Breathing exercises
 - Transfer training



PT Management :

- **Flexibility**
 - Active assistive movement
 - Passive movement
 - PNF with resistance
 - PNF Stretching (Contract Relax)
 - Yoga Poses
 - Unilateral bridging
 - B/L bridging
 - High kneeling with pelvis translation
 - Positioning
 - Resistive exercises
 - MRP



PT Management :

- Flexibility

- Sit to stand
- Weight bearing exercises
- Frenkel exercises
- Mobilizing facial muscles
- Backward stepping can be used to strengthen hip and spinal extensors and promote upright postures
- Patient with Parkinson experiences a high incident of falling. So patient should be taught how to get up after a fall.
- Quadruped creeping should be taught so that the patient is able to move to nearby support.

PT Management :

- **Balance Training**
 - Emphasize practice of dynamic stability task
 - Weight shifting
 - Alternate U/L weight bearing
 - Reaching
 - Axial rotation of Head and Trunk
 - Axial rotation with reaching
 - Quadruped position
 - Kneeling
 - Half Kneeling
 - Changing arm position like Arms folded cross chest, Touching the opposite shoulder etc

PT Management :

- **Balance Training**
 - Changing foot position
 - Overhead arm clapping
 - Heel raises
 - Toe standing
 - Toe walking
 - Partial wall squats
 - Chair raises
 - Single limb standing
 - Side kick & Back kick



PT Management :

- **Gait Training**
 - Verbal command
 - Visual and auditory cues
 - Task specific training will be more beneficial
- **Pulmonary rehab**
 - Major complication in PD are :
 - Upper respiratory obstruction
 - Restrictive disorder
 - Aspiration pneumonia
 - Diaphragmatic breathing
 - Postural drainage
 - Vibration and Shaking

PT Management :

- **Home Exercise Plan (HEP)**
 - Make sure the patient understood about the importance of self exercises
 - Early morning warmups
 - Self stretching and Strengthening
 - Hanging by hand using overhead bar – Prevent bending and kyphosis
 - Corner wall stretches



THANK YOU